

Course Description and Syllabus

SURV 617 (JPSM) / SURVMETH 687 (MPSDS)

Applications of Statistical Modeling

Joint Program in Survey Methodology, University of Maryland

Program in Survey and Data Science, University of Michigan

FALL, 2025

Abstract: *Applications of Statistical Modeling*, designed and required for students on all three tracks of the two programs in survey and data science, will provide students with exposure to applications of more advanced statistical modeling tools for both substantive and methodological investigations that are not fully covered in other MPSDS or JPSM courses. Modeling techniques to be covered include multilevel and marginal modeling techniques for clustered or longitudinal data (with applications to methodological studies of interviewer effects and modeling trends in the Health and Retirement Study), structural equation modeling (with an application of latent class models to methodological studies of measurement error), and machine learning algorithms (with an application to prediction of response propensity). Discussions and examples of each modeling technique will be supplemented with methods for appropriately handling complex sample designs when fitting the models. The class will focus on essential concepts, practical applications, and software, rather than extensive theoretical discussions.

Instructors: Brady T. West, Ph.D.; John Kubale, Ph.D.

Course: SURV 617 (JPSM) / SURVMETH 687 (MPSDS)
Dates: August 26 – December 9, 2025
Lectures: Tuesdays, 9:30am – 12:00pm (with the exception of December 9)
Locations: University of Michigan, Perry G300
University of Maryland, Lefrak 2208

Zoom Information:

<https://umich.zoom.us/j/92249807972>

Meeting ID: 922 4980 7972

Passcode: 041225

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Course Description

This course will introduce students to applications of more advanced statistical modeling tools for various substantive and methodological survey research investigations. The modeling techniques that will be covered in this course are typically given only brief, introductory overviews in other MPSDS and JPSM courses, and the course is therefore designed to provide students on all three tracks with more in-depth exposure to practical applications of the various models. In terms of learning outcomes, **students should expect to gain practical knowledge with regard to real-world applications of these various models, in addition to a thorough understanding of more advanced methods for fitting, interpreting, and diagnosing complex statistical models.**

The first general class of modeling tools covered in the course will be **multilevel and marginal models for clustered and longitudinal data**. Multilevel models are unique in that they include both *random effects* and *fixed effects*, enabling additional inferences about the variance in model coefficients between randomly sampled clusters or individuals at higher levels of a multilevel data hierarchy. They are also extremely flexible in their ability to effectively describe complex relationships in clustered or longitudinal data sets. In survey methodology, we are frequently interested in the effects that human interviewers have on the survey measurement process, and multilevel models provide natural tools for describing these effects, given that we can view the interviewers working on a given survey as being randomly sampled from a larger pool of hypothetical interviewers that could have worked on the project. After introducing general and essential statistical concepts related to multilevel modeling, we will then focus on a specific application of multilevel modeling to the study of interviewer effects in the European Social Survey, or ESS (focusing on a Belgian sample specifically). We will walk through a full example of specifying a model, using existing software to fit the model, and then interpreting and describing the results of that model in an academic report. Examples of various software procedures that can be used to fit different types of multilevel models will be covered as well, and this portion of the course will conclude with a discussion of how complex sample design features should be accounted for when fitting these types of multilevel models.

We will then turn to alternative models for longitudinal data. We will begin with a discussion of how multilevel models can be used to estimate trajectories in survey outcomes of interest in panel surveys, and estimate variance between subjects in terms of their trajectories. We will then consider alternative marginal modeling techniques for longitudinal data, including *marginal linear models with correlated errors*, and *generalized estimating equations (GEE)*. These alternative techniques will all be applied to longitudinal data from the Health and Retirement Study (HRS), enabling comparisons of results and the inferences that are possible when using the techniques. Techniques for accommodating complex sample design features when fitting models to longitudinal data (including correct handling of time-invariant and time-varying survey weights) will also be covered at the conclusion of this portion of the class.

The second general class of modeling tools covered in the course will be **structural equation models**. These models are generally defined by a combination of *measurement models*, which

describe latent (unobserved) constructs that are not directly measured in a survey but rather indicated by multiple survey items, and *structural models*, which describe the causal relationships between latent constructs. *Path models* also fall under this general class of modeling tools, and describe causal relationships between variables that have been fully observed in a survey. Survey methodologists typically use structural equation models to handle problems with measurement error in survey items that supposedly measure the same construct. After introducing the general and essential concepts related to fitting structural equation models, we will consider an application of structural equation modeling to the problem of measurement error in survey items, using *latent class analysis*, and test hypotheses about causal relationships between variables of interest in a panel survey focusing on substance use behaviors (the National Epidemiological Survey on Alcohol and Related Conditions, or NESARC). Examples of various software procedures that can be used to fit these models will be covered, and methods for fitting these models when analyzing data from complex samples will be introduced as well.

The final general class of modeling tools covered in the course will be **machine learning algorithms**. We will cover classification trees, random forests, and gradient boosting trees, with a focus on basic concepts, applications, and comparisons from the machine learning perspective. These models are generally used as a type of unsupervised learning tool and can be used to uncover complex interactions between predictor variables that could be used to classify cases in large data sets (e.g., “Big Data”). We will apply this modeling technique to the prediction of response propensity in the second wave of the NESARC (using Wave 1 covariates to build the classification tree). Response propensity modeling is often used for nonresponse adjustment purposes in large surveys, but logistic regression is frequently used to estimate these response propensities. Classification trees offer the advantage of identifying complex interactions between variables that strongly influence response propensity, and such interactions may be hard to identify using standard logistic regression modeling techniques. Random forests and gradient boosting are both extensions of classification trees combined with ensemble techniques. Methods for accounting for complex sample designs when building classification trees will also be covered briefly.

A variety of software tools will be illustrated throughout the class when fitting the models, including procedures from R, Stata, SAS, and Mplus; there will not be a focus on any one software package, although one package may be favored depending on a poll of the students. Examples of software code (or menu steps) that can be used to fit the models in a given software package will be provided regularly, in addition to annotated output from the software.

Prerequisites

The prerequisites for SURV 617 / SURVMETH 687 include SURV 615 / SURVMETH 685 (**Statistical Methods I**), SURV 616 / SURVMETH 686 (**Statistical Methods II**), SURV 400 / SURVMETH 600 (**Fundamentals of Survey Methodology**) and SURV 625 / SURVMETH 625 (**Methods of Survey Sampling / Applied Sampling**), or equivalents of these four courses (e.g., Applied Statistics I and II, Introduction to Survey Methodology / Survey Research Methods, and Applied Sampling or Sampling Theory. Many Survey and Data Science students will likely be taking SURV 701 / SURVMETH 701 (Analysis of Complex Sample Survey Data) concurrently. Permission of the instructor is also possible given adequate prior course work in graduate-level applied statistics and strong student interest in survey methodology. The course will be presented at a moderately advanced statistical level, and will assume that **students are very familiar with commonly applied statistical methods (including multiple regression and logistic regression), maximum likelihood estimation principles, applied sampling methods, and**

hypothesis testing concepts. Students should be familiar with at least one software tool, such as R, Stata, SAS, or Mplus.

Course Format

Students in the class will meet on a weekly basis in the respective classrooms at Ann Arbor and College Park. **In-person attendance is required; individual attendance by Zoom on a given day must be approved by the instructors.** Class time will be used for a combination of lectures and discussions of examples and analysis projects. Lecture notes and examples will be presented on PowerPoint slides, and copies of these materials will also be available to each student on the course website (Canvas). Software packages demonstrated in a “live” fashion during class sessions will include R, Stata, SAS, and Mplus. Questions are welcomed during lectures (see more below), and off-line discussions are encouraged.

Class Readings

There is no single textbook for this course, given the variety of topics covered. We recommend the following two books for additional readings on statistical modeling.

Regression and Other Stories, Andrew Gelman, Jennifer Hill, Aki Vehtari, 2020, Cambridge press. (Accessible at <https://avehtari.github.io/ROS-Examples/>)

Data Analysis Using Regression and Multilevel/Hierarchical Models, Andrew Gelman and Jennifer Hill, 2006, Cambridge press.

Students will generally be assigned a handful of readings on a weekly basis, related to the modeling topics that will be covered in lecture in the following week. **There will be assigned readings for the first week of class.** Students with the aforementioned pre-requisites should be able to process the selected readings without much difficulty, and are welcome to email the instructors or **post questions on Canvas for discussion**, should any questions about the readings arise. The readings have been specifically selected to be more practical in nature, rather than theoretical, given the focus of the course.

Students are required and strongly encouraged to finish all assigned readings prior to the lecture for which they have been assigned. While this will not be checked specifically via quizzes or required email questions about the readings, class participation will be monitored carefully. **Frequent in-class participation noted by the instructors will be used to adjust the final grades for each analysis project in an upward direction, while frequent failure to participate will be used to adjust the final grades in a downward direction.** The instructors will be in constant communication about any perceived lack of participation, to ensure that this is not due to a failure to understand the course material.

Grading

There will be a two-hour, in-class, cumulative, **open book / open notes** midterm examination on **Tuesday, October 7**. *(If this examination time conflicts with other regularly scheduled examination times for University of Michigan or University of Maryland students, students must inform the instructors as soon as possible.)*

Final grades will be a weighted composite of the average score on the three homework assignments (30%; see below), the midterm exam (30%), and a final analysis project report (40%). Additional details about the analysis project and the grading criteria that will be used can be found below. The final grades will not be based on any kind of curve, and will use a standard grading scale (97-100% = A+, 93-96% = A, 90-92% = A-, 88-89% = B+, 83-87% = B, 80-82% = B-, <79% = C).

Homework

Three homework assignments are planned. Each is to be turned in by the beginning of the class session when due (see syllabus below), and each will receive equal weight for the final grade.

Homework assignments correspond with the course units and are designed to aid in skill development. Assignments will be graded check-plus (100 points), check (90), check-minus (80), and not submitted (0), and will ordinarily be marked and returned before the next class session. Students may request permission to submit homework late via email to the instructors, but the request must be no later than one hour before the homework is due. Permission is not guaranteed, although typically granted. If a late submission is granted, there will be an agreed-upon date and time when the late assignment must be submitted; such assignments will be graded using the specified marking system. If homework is submitted late without prior permission, scores will be check-plus (70), check (60), and check-minus (50).

Analysis Projects

Project Teams. Students will form teams of 2-4 for the analysis project. **All students on a team will be 100% responsible for careful proofreading and editing of the final project report.** English grammar and clear technical writing will be an important part of the overall grade on each project; please see the PDF file **styletips.pdf** on Canvas (in the “**Miscellaneous**” file folder), courtesy of Dr. Rod Little, for importance advice. Each team will be able to decide whether they would like to either use the data set that has been discussed in class for a given topic or use a data set of their own choosing with the required variables present (e.g., interviewer / cluster ID codes). Students working on the same team are strongly encouraged to schedule a weekly meeting time **outside of lecture** to make progress on the final project. The grading will consist of two parts: 1) a team project report score (max 80 points); and 2) an individual evaluation score (20 points). Each student will evaluate the contributions of each of the other team members to the team project report. The student’s teammate’s evaluation scores will be averaged to generate the individual student project score. Each student must complete an evaluation for the other students on the team; students not submitting evaluations will receive a score of zero for the individual component score.

Analysis Project Requirements. Each team will be responsible for working on the following tasks throughout the semester (maximum points on the final project for each task are indicated in parentheses):

1. *Research Question (5 points)*: Formulating a methodological or substantive research question that can be addressed using one of the modeling techniques discussed in the class, including 1-2 (maximum) important pieces of literature motivating the research question;
2. *Data Set (5 points)*: Briefly describing the data set and its structure, identifying variables in the data set that you will analyze (possibly including complex sample design features), and presenting simple (yet appropriate) descriptive statistics for each of the variables;
3. *Modeling Approach (20 points)*: Specifying and clearly defining an appropriate statistical model for analyzing the data and answering the research question (**being extremely careful with notation**), and describing the general modeling approach;
4. *Software Coding (10 points)*: Selecting software that you can use to fit the model and writing appropriate code that will compute the estimates of interest;
5. *Interpretation and Inference (30 points)*: Correctly interpreting the modeling results, assessing model diagnostics / quality of fit, and making appropriate inferences with regard to your research question(s); and
6. *Analysis Report (80 points total)*: Drafting a brief analysis report that describes Sections 1 through 5 above in clear detail (including model notation), **including overall conclusions (10 points)**, interpretation of results, and software code used (**clearly commented in a required Appendix**).

The final report should be submitted for grading through Canvas (only), no later than 5:00pm on the due date indicated in the syllabus. The reports will be graded with respect to: 1) Parts 1 through 6 above; 2) the clarity of the material presented; 3) the quality of the technical writing and the English grammar; and 4) correct interpretation of the results. Figures and tables can certainly be used to enhance the report and describe the data being analyzed or the results of interest. Analysis reports should be **double-spaced, using 12-point font, and no more than 10 pages long**, including all equations, tables and figures (tables and figures will **not** be allowed in appendices). This makes concise writing extremely important. **The software code provided in the Appendix should be clearly commented and easy to follow.** An example of a high-quality analysis project from a previous year is provided on Canvas (under **Files**) as a reference point. **There will be a 5% reduction in the score for a given analysis project for each day after the due date that it takes for a team to submit the project via Canvas.**

Accommodations for Students with Disabilities

University of Michigan

The University of Michigan recognizes disability as an integral part of diversity, equity, and inclusion and is committed to creating an accessible educational environment for students with disabilities.

Disability can include: mental health conditions, ADHD, learning disabilities, autism, chronic illness, physical conditions, sensory conditions, and more.

If you anticipate or are experiencing barriers based on disability or temporary injuries, Services for Students with Disabilities (SSD) (ssd.umich.edu) is the office that students work with to explore reasonable accommodations, tools, and resources.

- If **you are already connected** with SSD and have approved accommodations, please share your [letter through Accommodate](#) as soon as possible so that we can discuss how your accommodations will be implemented in this course. The sooner I know about your disability access needs, the more equipped I can be to facilitate accommodations. You should contact me and/or your Disability Access Coordinator with any accommodation questions or concerns.
- If **you have not connected** with SSD and anticipate or are experiencing a disability-related barrier, and would like to discuss accommodations and/or resources, please contact SSD by completing their initial information form (<https://ssd.umich.edu/article/how-get-connected>)
- If you have a **temporary medical injury/condition**, such as a broken arm, I may be able to help minimize classroom barriers. In situations where additional assistance is needed, you should contact the SSD, as noted above.

For more information, call 734-763-3000 or email ssdoffice@umich.edu.

University of Maryland

For information about Accessibility and Disability Service (ADS) on campus, visit this website: <https://www.counseling.umd.edu/ads/>.

Accessibility and Disability Service (ADS) Main Office:

Phone: [301.314.7682](tel:301.314.7682)

0106 Shoemaker Building

Dissup@umd.edu

Office Hours: Monday - Friday (8:30am to 4:30pm)

ADS Testing Office (for ADS exams):

Phone: [301.314.7217](tel:301.314.7217)

0118 Shoemaker Building

DSSTest@umd.edu

ADS Exam Hours: Monday - Friday (9am to 4pm)

Overview of ADS Services:

In order to receive services you must contact our office to register in person for services. Please call the office to set up an appointment to register with an ADS counselor. Contact the ADS office at [301.314.7682](tel:301.314.7682).

Academic Conduct

Please familiarize yourself with the respective campus policies on academic integrity:

- a. UMD: <https://studentconduct.umd.edu/what-academic-misconduct>

- b. UMICH: <https://guides.lib.umich.edu/academicintegrity>

****Given the objectives of this class, there will be a zero-tolerance policy for any form of plagiarism. Any evidence of plagiarism will likely lead to a failing grade for the class and immediate removal of a student from the class.**

Reporting of Sexual Harassment:

Dr. West is an **individual** at the University of Michigan with **reporting obligations (IRO)**. Please do not hesitate to report any incidents of inappropriate sexual conduct to Dr. West as soon as possible.

Use of AI Tools:

While the instructors cannot fully prevent students from using generative AI tools (like Chat GPT) to assist with writing and development of code, and the instructors encourage the use of these tools to improve the clarity of their writing and the efficiency of their code, students should not rely on these tools to write text for them, perform the required data analysis for their projects, or meet any of the other course objectives. Students using AI tools for course assignments are requested to provide a written statement acknowledging their use and clarifying for what specific aspects of the projects and assignments the tools were used. Both instructors will be able to readily detect inconsistencies in writing suggesting that large portions of the analysis project were written with generative AI, and if this is not formally acknowledged, substantial reductions in grades may occur.

Course Schedule / Assigned Readings / Deadlines

All assigned readings need to be completed prior to the start of the indicated class, and topics from these readings will be discussed in class. **The order of the topics and assigned readings are subject to change.**

Class Date	Topic / Deadline	Assigned Readings
August 26	Course overview. Review of ordinal linear regression and logistic regression models.	1. Syllabus 2. Regression and Other Stories, Gelman, Hill and Vehtari, Ch. 10, 11 and 13 (see web link above)
September 2	Multilevel modeling background and concepts. Introduction of European Social Survey (ESS) data. ** HW 1 assigned: multilevel models	1. Gill and Womack (2013) 2. Merlo et al. (2005)
September 9	Software for fitting multilevel models.	1. Galecki and West (2013) 2. West and Galecki (2011).
September 16	Interviewer effects: A review.	1. West et al. (2013)

	Multilevel modeling application: Interviewer effects in ESS.	2. O’Muircheartaigh and Campanelli (1998)
September 23	Alternative statistical models for longitudinal data: An overview. Software for fitting models to longitudinal data. **HW 1 DUE: multilevel models	1. Ballinger (2004) 2. Steele (2008)
September 30	Application: Alternative approaches to fitting growth curve models to HRS data.	1. Kreuter and Muthen (2008) 2. Hubbard et al. (2010) 3. Twisk (2004)
October 7	Midterm Examination	
October 14	No Class	
October 21	Accounting for complex sample designs when fitting multilevel/marginal models. ** HW 2 assigned: marginal models	1. Carle (2009) 2. Rabe-Hesketh and Skrondal (2006) 3. West et al. (2025) 4. Veiga et al. (2014) 5. Thompson (2015)
October 28	Structural equation models (SEMs): Overview.	1. Hox and Bechger (1998)
November 4	Multiple Group Analysis (MGA) and Latent Class Analysis (LCA). Software for LCA. ** HW 2 DUE: marginal models; HW 3 assigned: structural equation models	1. Lanza et al. (2007) 2. Kreuter, Yan, and Tourangeau (2008)
November 11	SEM application: Applying LCA to data from the National Epidemiologic Survey of Alcohol and Related Conditions (NESARC)	1. McCabe and Cranford (2012) 2. Biemer and Wiesen (2002)
November 18	Accounting for complex sample designs when fitting structural equation models. ** HW 3 DUE: structural equation models	1. Stapleton (2006) 2. Oberski (2014)
November 25	Classification trees: Overview and software **Preliminary draft of final report DUE	1. Lemon et al. (2003) 2. Lewis (2000) 3. Ledolter (2013)

December 2	Classification tree application: Prediction of response propensity in Wave 2 of the NESARC.	1. Wun et al. (2007) 2. Tollenaar and van der Heijden (2013)
December 9	**Final project report DUE at 5:00pm (Canvas)	